

LOSSIEMOUTH, United Kingdom

WMO: 030680

Lat: 57.7114N

Lon: 3.3233W

Elev: 7

StdP: 101.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.9	(c) -2.9	(d) -6.8	(e) 2.1	(f) -3.2	(g) -5.2	(h) 2.4	(i) -0.9	(j) 17.8	(k) 5.3	(l) 15.5	(m) 5.4	(n) 2.1	(o) 240	(p) 0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.8	(c) 22.2	(d) 16.4	(e) 20.4	(f) 15.6	(g) 19.0	(h) 14.8	(i) 17.3	(j) 20.9	(k) 16.4	(l) 19.4	(m) 15.5	(n) 18.1	(o) 6.2	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.3	18.9	15.0	10.6	17.8	14.2	10.1	16.8	48.8	20.9	46.0	19.5	43.4	18.2	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.8	12.2	10.9	-7.9	25.9	3.4	2.3	-10.4	27.6	-12.4	29.0	-14.3	30.3	-16.7	31.9		
			-8.4	19.4	3.4	1.5	-10.8	20.4	-12.8	21.3	-14.7	22.1	-17.2	23.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.8	4.1	5.5	7.6	9.8	12.4	14.5	14.3	12.5	9.5	6.3	3.9
	DBStd	4.75	2.82	2.96	2.80	2.80	2.75	2.33	2.38	2.20	2.32	2.70	2.92	3.61
	HDD10.0	974	191	166	143	82	37	4	0	1	4	42	114	191
	HDD18.3	3517	449	398	399	322	264	178	121	126	177	274	360	447
	CDD10.0	506	0	1	2	10	31	77	141	134	78	26	4	1
	CDD18.3	6	0	0	0	0	0	0	3	2	1	0	0	0
	CDH23.3	28	0	0	0	0	2	2	16	9	1	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0
Wind	WSAvg	5.3	5.9	6.0	6.0	5.5	4.9	4.8	4.4	4.5	5.0	5.4	5.6	5.5
Precipitation	PrecAvg	738	56	52	48	49	54	67	65	70	72	82	66	58
	PrecMax	900	107	104	84	138	120	127	130	183	303	141	106	95
	PrecMin	554	12	21	15	17	16	30	25	18	26	24	17	15
	PrecStd	82	26	21	16	27	23	28	32	36	52	33	23	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.6	14.9	18.2	21.3	22.4	25.6	24.5	21.8	18.1	15.0	13.6
		MCWB	9.9	10.4	10.1	12.1	13.9	15.9	17.8	17.7	16.2	14.1	12.2	10.8
	2%	DB	10.9	12.1	12.8	15.7	18.1	19.9	22.3	21.6	19.5	16.3	13.1	12.2
		MCWB	8.6	9.4	9.2	11.0	13.1	14.9	16.8	16.5	15.1	12.8	10.8	9.8
	5%	DB	9.6	10.5	11.4	14.0	16.1	18.2	20.4	19.8	17.9	15.0	11.9	10.6
		MCWB	7.6	8.1	8.3	9.9	11.8	13.8	15.8	15.3	14.0	12.0	9.8	8.6
	10%	DB	8.2	8.9	10.2	12.4	14.4	16.7	18.8	18.2	16.5	13.8	10.8	9.2
		MCWB	6.5	6.8	7.5	9.0	10.8	12.9	15.0	14.5	13.1	11.2	9.0	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.0	11.0	12.9	15.4	17.3	19.1	18.4	16.9	14.5	12.5	11.1
		MCDB	12.1	13.0	14.1	16.9	18.9	20.8	23.7	22.8	20.8	17.3	14.6	13.1
	2%	WB	8.8	9.5	9.6	11.5	13.6	15.4	17.6	17.1	15.6	13.4	11.1	10.0
		MCDB	10.6	11.7	12.3	15.1	17.1	18.9	21.0	20.5	18.6	15.6	12.8	12.0
	5%	WB	7.7	8.3	8.6	10.3	12.4	14.4	16.6	16.2	14.6	12.5	10.1	8.7
		MCDB	9.3	10.4	11.0	13.4	15.5	17.3	19.5	19.0	17.0	14.5	11.6	10.4
	10%	WB	6.6	6.9	7.6	9.2	11.3	13.4	15.6	15.2	13.7	11.6	9.2	7.5
		MCDB	8.1	8.7	9.8	11.8	13.9	16.1	18.1	17.4	15.8	13.4	10.6	9.1
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.1	6.6	7.0	7.1	6.5	6.8	6.9	6.8	6.5	5.7	5.7
		MCDBR	6.4	7.2	8.4	9.6	10.3	9.5	9.8	9.5	8.9	7.9	6.8	6.8
		MCWBR	5.3	5.6	5.7	5.8	5.9	5.5	5.4	5.4	5.5	5.6	5.6	5.7
	5% WB	MCDBR	6.2	7.0	7.9	9.0	9.4	8.7	8.8	8.7	8.2	7.3	6.5	6.7
		MCWBR	5.3	5.6	5.8	5.7	5.8	5.4	5.3	5.4	5.6	5.7	5.6	5.8
Clear-Sky Solar Irradiance	taub		0.307	0.332	0.355	0.375	0.374	0.372	0.376	0.364	0.349	0.335	0.309	0.292
	taud		2.250	2.346	2.345	2.325	2.373	2.408	2.405	2.447	2.489	2.446	2.298	2.180
	Ebn at Noon		540	707	792	835	861	867	854	841	799	700	535	430
	Edh at Noon		50	70	91	108	110	107	106	95	79	63	47	39
All-Sky Solar Radiation	RadAvg		0.42	1.05	2.10	3.46	4.67	4.59	4.44	3.50	2.44	1.28	0.55	0.29
	RadStd		0.05	0.10	0.22	0.33	0.25	0.37	0.36	0.22	0.22	0.11	0.04	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page